CM3191: How to Check NMR Spectrometer Frequency in MestReNova

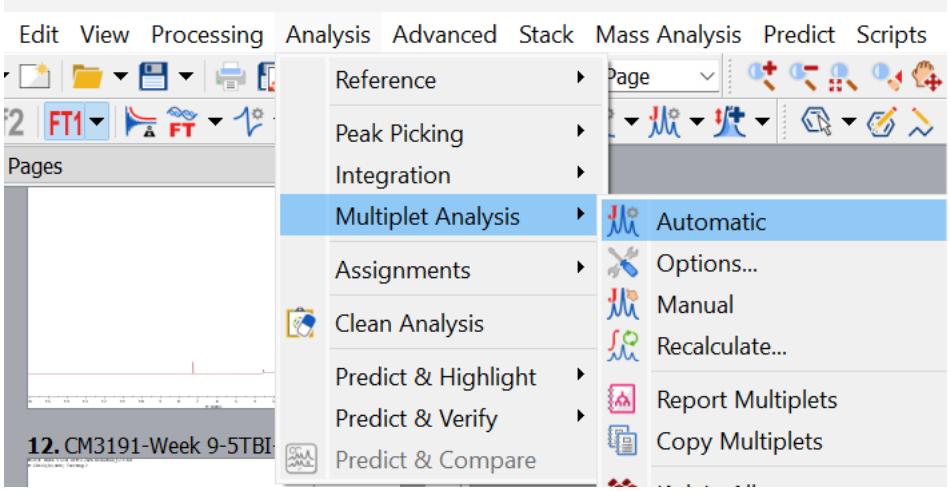
Samples are typically acquired on either a 400 MHz or 500 MHz instrument. Use either of the methods below to check which spectrometer frequency was used for your dataset.

Method 1: Via Multiplet Analysis Report (Fastest)

1. Open your spectrum in MestReNova.
2. Go to the top menu and select:

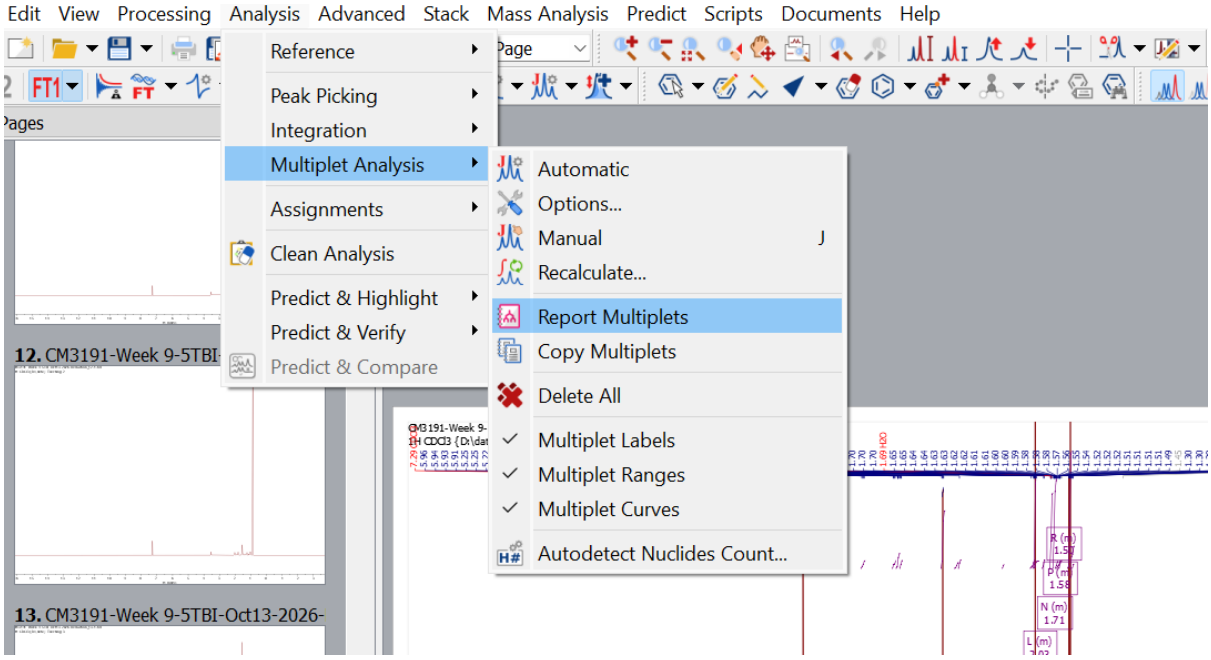
Analysis → Multiplet Analysis → Automatic

(or press the corresponding icon on the toolbar).

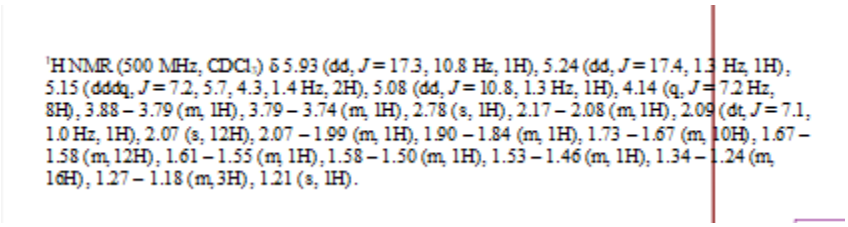


3. Generate the report by selecting:

Analysis → Multiplet Analysis → Report Multiplets.



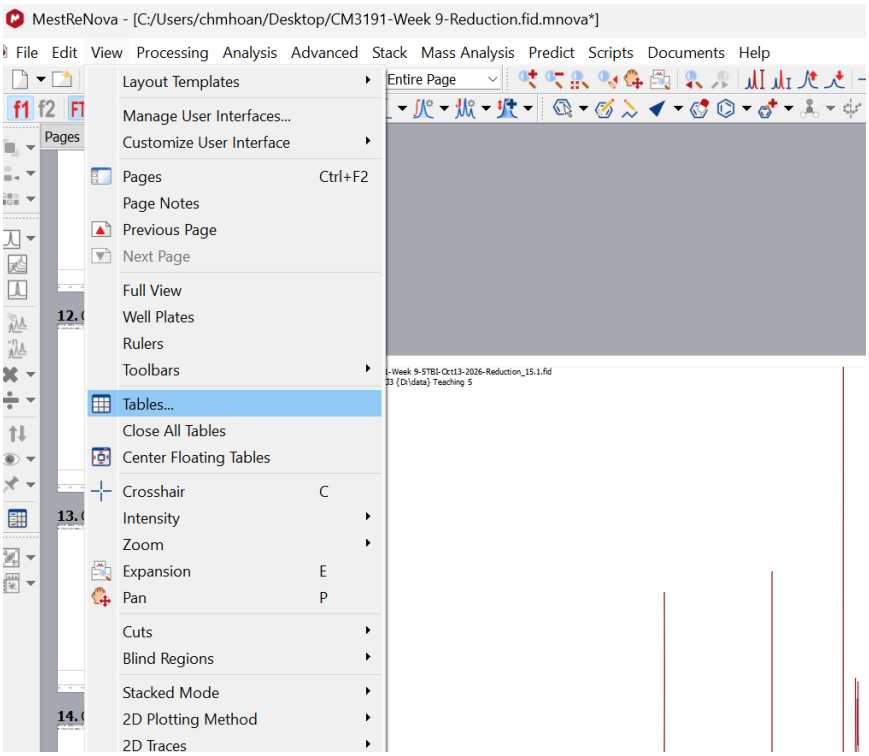
4. A text box formatted for publication/reports will appear directly on your spectrum canvas.
    - Identify the frequency enclosed in parentheses at the start of the report (e.g., ^{1}H NMR (500 MHz, $CDCl_3$)... indicates a 500 MHz instrument).



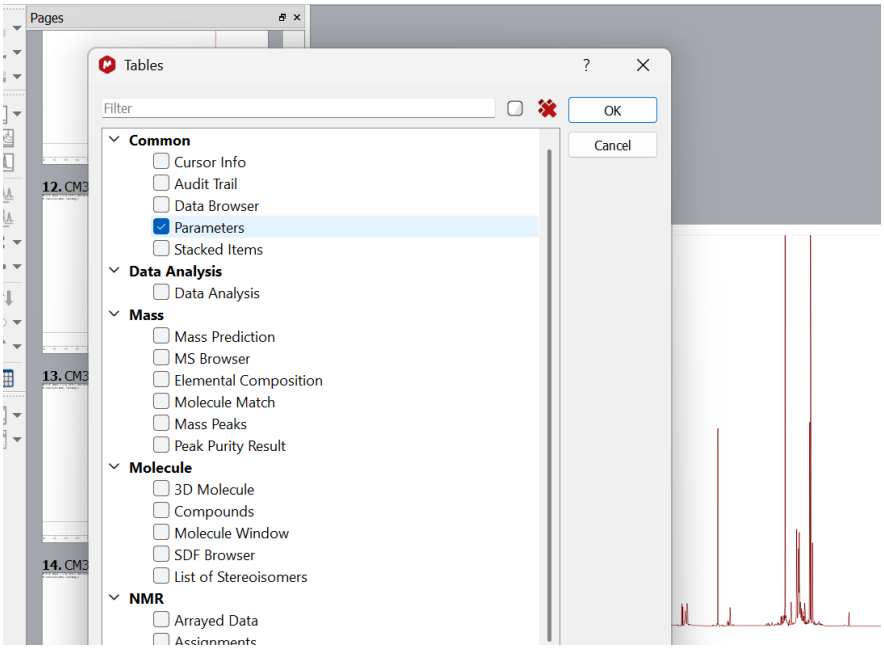
Method 2: Via Acquisition Parameters Table (Direct Readout)

1. Open your spectrum in MestReNova.
2. In the top menu bar, select:

View → Tables...



3. In the pop-up dialog under the Common section, check the box for Parameters, then click OK.



4. A Parameters table will open:
    - Scroll down to find the row labeled Spectrometer Frequency.
    - The value column displays the exact frequency in MHz (e.g., 500.29 corresponds to a 500 MHz instrument; ~400.xx corresponds to a 400 MHz instrument).

	Parameter	Value
10	Temperature	298.0
11	Pulse Sequence	zg30
12	Experiment	1D
13	Probe	Z144569_0005
14	Number of Scans	8
15	Receiver Gain	101.0
16	Relaxation Delay	1.0000
17	Pulse Width	10.0000
18	Presaturation Frequency	
19	Acquisition Time	3.2768
20	Acquisition Date	2026-10-13T09:22:52
21	Modification Date	2026-10-13T10:28:28
22	Class	
23	Method Name	C:/Users/u2103/AppData/Local/Temp/icnklzwuQPG1f5QKN5fMmA/CM3191-Week 9-5TBI-Oct13-2026-Reduction_15/1/pdata/1/procs
24	Spectrometer Frequency	500.29
25	Spectral Width	10000.0

	Parameter	Value
1	Spectrometer Frequency	500.29